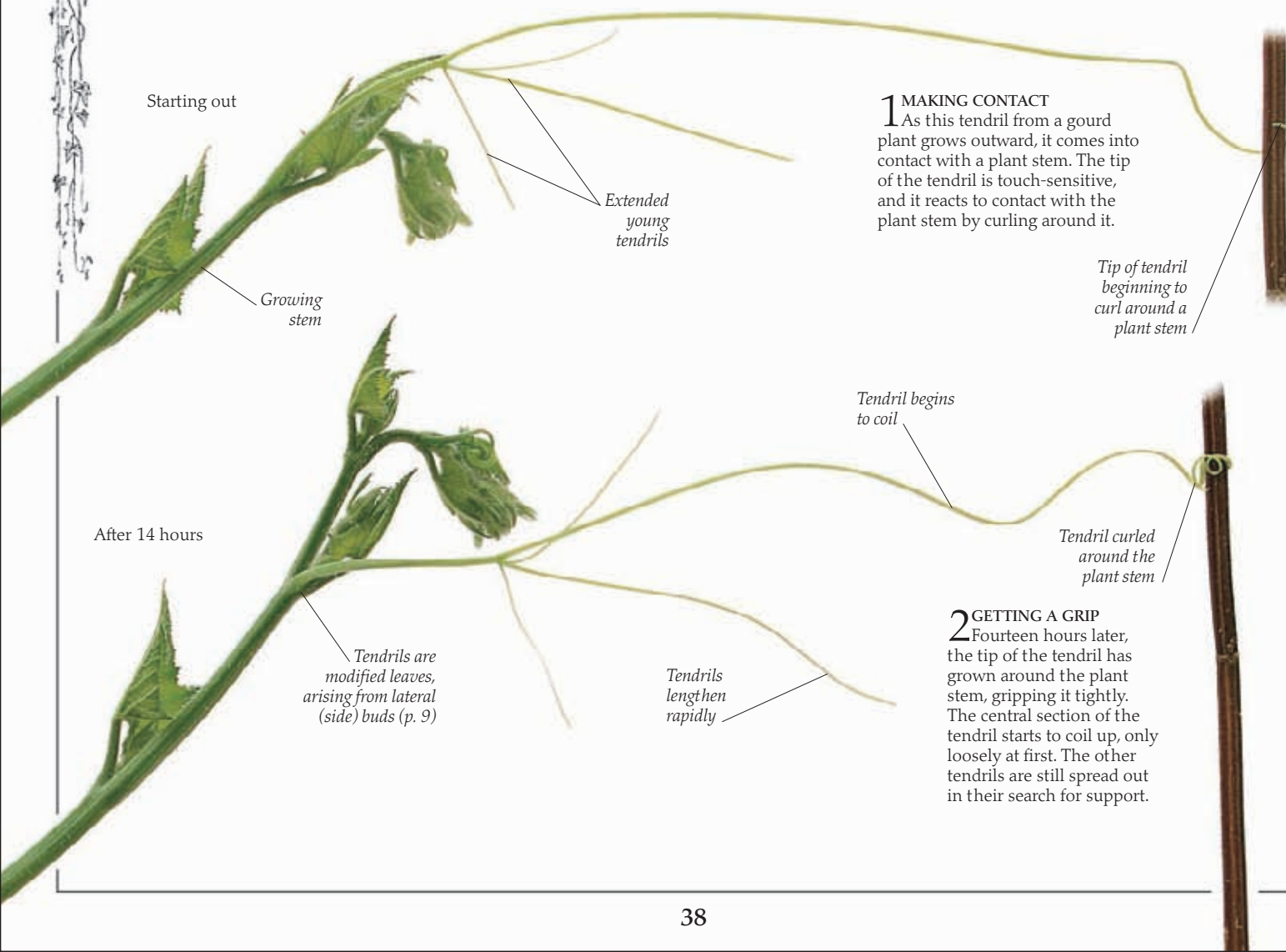


Creepers and climbers

WHEREVER THERE IS MOISTURE AND WARMTH, plants struggle against each other for light. The tallest plant usually gets the greatest share, but it also has to spend the most energy in growing a strong stem, or a tree trunk, to hold up its leaves. But there are some plants—epiphytes (pp. 46–47) and climbers—that take a shortcut to the top. They take advantage of other plants and even buildings to get a place in the light with much less effort. Epiphytes may grow on the trunks or upper branches of trees and are lifted up with them as they grow. These plants do not have roots on the ground and are able to absorb all the water they need from the air and rainwater. Climbers need supports. Some twine themselves around a plant, while others put out touch-sensitive feelers, or tendrils, that curl around the support when they come into contact with it. A third group of climbers raise themselves by means of stiff side branches, prickles, roots, or hairs.



GROWING IN A SPIRAL
Plants that grow in a spiral will twist in a set direction. Scarlett runner plants always twist in a clockwise direction, a detail noticed by the artist who made this 16th-century woodcut of a bean plant climbing up a stick.



1 MAKING CONTACT

As this tendril from a gourd plant grows outward, it comes into contact with a plant stem. The tip of the tendril is touch-sensitive, and it reacts to contact with the plant stem by curling around it.

Tip of tendril beginning to curl around a plant stem

Tendril begins to coil

Tendril curled around the plant stem

2 GETTING A GRIP

Fourteen hours later, the tip of the tendril has grown around the plant stem, gripping it tightly. The central section of the tendril starts to coil up, only loosely at first. The other tendrils are still spread out in their search for support.